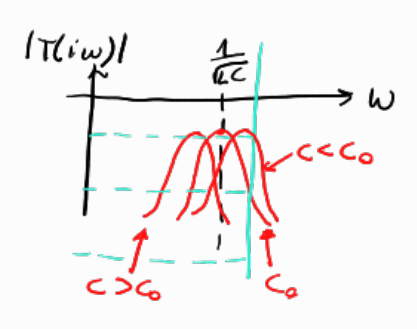
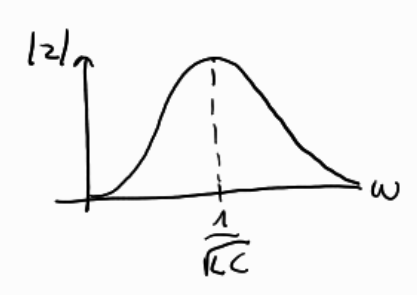
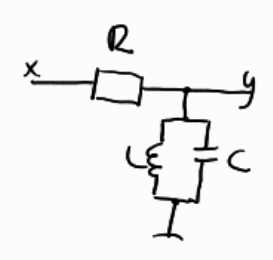
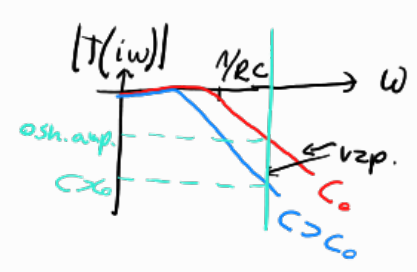
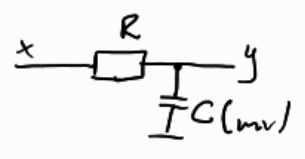
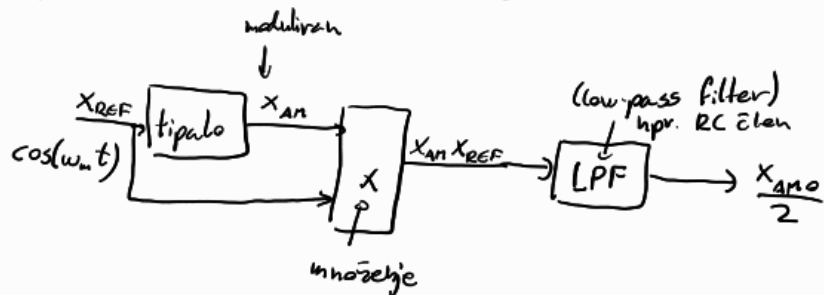


$$y = x \frac{R_2(mv)}{R_1 + R_2(mv)} ; R_1 \gg R_2(mv)$$

$$\approx x \frac{R_2(mv)}{R_1}$$



VEZJE ZA SINHRONO DEMODULACIJO

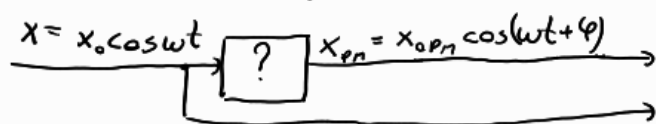


$$x_{an} = x_{amo} \cos \omega_n t$$

$$x_{an} x_{ref} = x_{amo} \cos^2 \omega_n t = \frac{x_{amo}}{2} (1 + \cos 2\omega_n t)$$

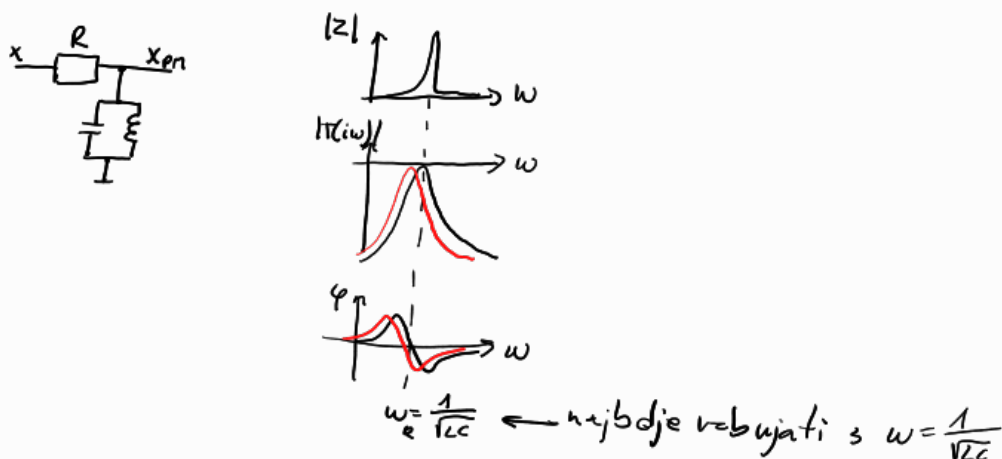
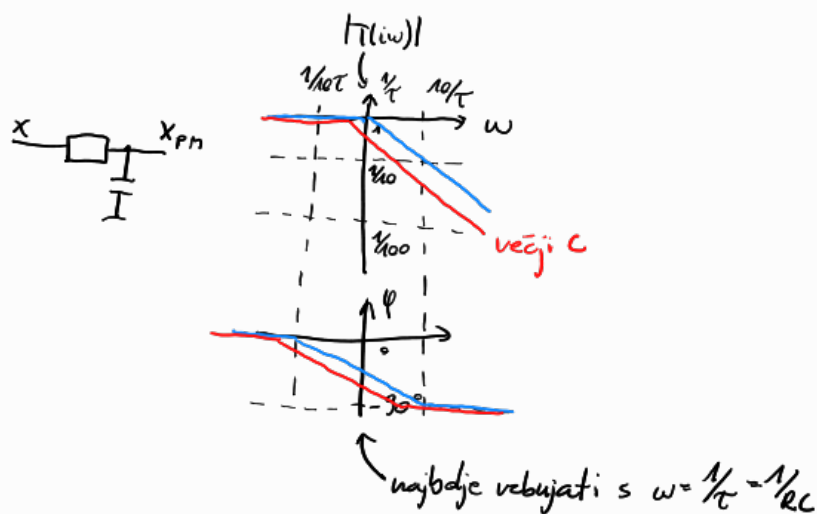
↓
se želimo znebiti

Fazna modulacija (PM)

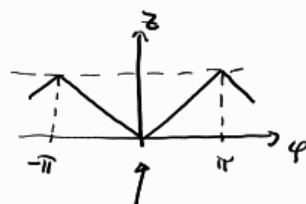
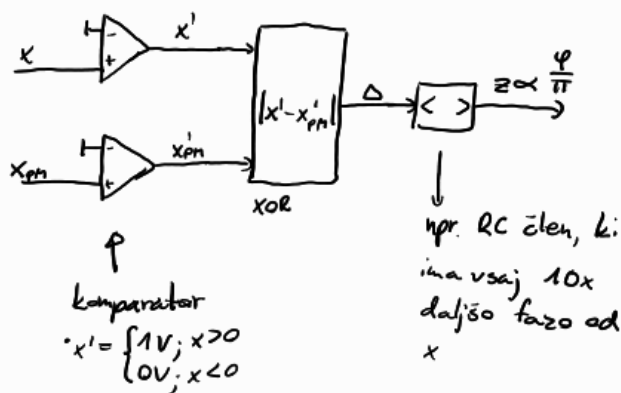


$$\varphi = \varphi_{n0} + \varphi_{m1} \cos \omega_n t$$

$$\rightarrow x_{pm} = x_{opm} \cos(\omega t + \varphi_{n0} + \cos \omega_n t) \rightarrow \text{reševanje s simulacijo}$$



Demodulacija faze



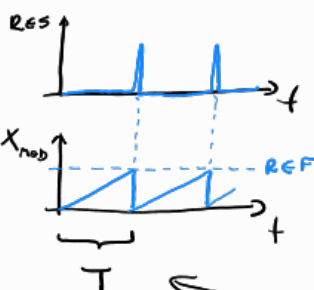
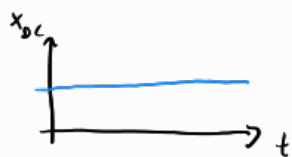
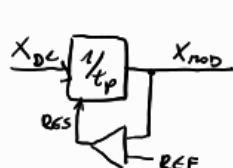
če želimo videti, ali je $\phi \neq 0$, je bolje ϕ_0 nastaviti na $\pi/2$

Frekvenčna modulacija (FM)

$$x_{FM} = x_o \cos(\omega t + \phi)$$

$$f = \frac{\omega}{2\pi} = f_{m0} + \Delta f_m \cos(2\pi f_m t)$$

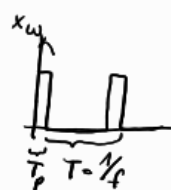
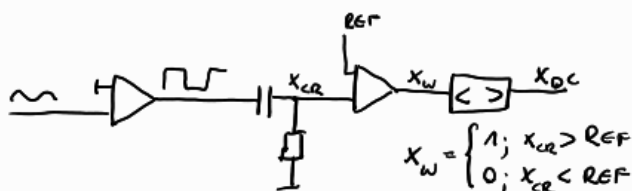
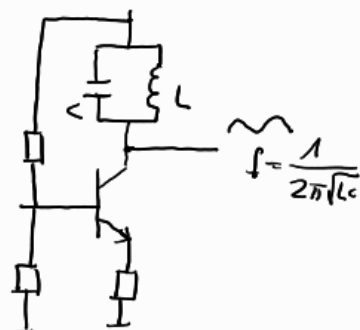
$$x_{FM} = x_o \cos\left(2\pi \int_0^t f(\tau) d\tau\right) = x_o \cos\left(2\pi f_{m0} t + \frac{\Delta}{f_m} \sin(\omega_m t)\right)$$



$$\frac{1}{RC} \int_0^T x_{DC} dt = REF$$

$$\frac{x_{DC}}{RC} T = REF$$

$$f = \frac{1}{T} = \frac{x_{DC}}{REF} \frac{1}{RC}$$



$$x_{DC} = T_p f$$